

KEREN SUNADA MALUGURI

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SUMMARY

Recent graduate in Electrical & Electronics Engineering with a solid academic background and practical experience in substation operations and electric vehicle design. Passionate about integrating theoretical knowledge with emerging technologies to foster advancements in electric mobility. Seeking an entry level opportunity to contribute innovative solutions in my field while continually improving my skills and expertise.

EDUCATION

Jawaharlal Nehru Technological Gurajada Vizianagaram (JNTGV)	(2022-2025)
<i>B.Tech (Bachelor of Technology) in EEE.</i>	CGPA: 7.42
Damacharla Anjaneyulu Govt Polytechnic, Ongole	(2019-2022)
<i>Diploma in Electrical and Electronics</i>	Percentage: 76.67

SKILLS

Programming Languages: Python, C

Tools: MATLAB Simulink, SAM, Microsoft Office (Word, PowerPoint, Excel)

Core Skills: Networks, Control Systems, DC Machines and Transformers, Electronics, Electric Vehicle (EV)

Soft Skills: Communication, Event management, Documentation and Reports preparation, Time management.

PROJECTS

Simulation of Solar Fed Buck Converter for DC Nano Applications (B. Tech Major Project)

- Designed a DC nano grid system integrated with solar photovoltaic (PV) generation for efficient DC power distribution.
 - Implemented a buck-boost DC–DC converter to obtain regulated 12V and 24V DC outputs for various DC loads.
 - Developed and applied Perturb and Observe (P&O) based Maximum Power Point Tracking (MPPT) algorithm to maximize solar energy harvesting.
 - Reduced power conversion losses by eliminating multiple AC–DC conversion stages.
 - Simulated and validated system performance using MATLAB/Simulink under varying solar conditions.
 - Ensured reliable power supply for DC loads such as mobile chargers and battery energy storage systems.
 - Improved overall energy efficiency and system stability for distributed energy resource applications.
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INTERNSHIP

33/11 kv Substation, Ongole, (6 Months)

OCT 2021 - Mar 2022

- Learned about power distribution in Ongole substation.
 - Observed various power equipment including transformers, circuit breakers, lightning arrestors, isolators, and other switchgear equipment at different voltage levels.
 - Gained understanding of the functions and importance of each equipment.
 - Learned about the protection scheme in Ongole for external and internal faults.
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CERTIFICATIONS & Activities

- EDX Certification in Solar PV cells
- IoT and Robotics Workshop
- Python programming basics

Activity: Coordinated two-day national level workshop on **GENERATIVE AI** an outreach workshop of **Technex'24, IIT(BHU) Varanasi**.

LANGUAGES

- English (Proficient)
- Telugu (Native)
- Hindi (Conversational)